
Agentic AI Marketing Decision Lab: Predictive Modeling and Business Decision Support

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Goals this week:

Plan data agent

Plan and validate the Data Agent so it works for all three prediction questions, not just Q1.

Models – R&D

Define the target, data scope, and method for each model: response probability, expected value, and true uplift.

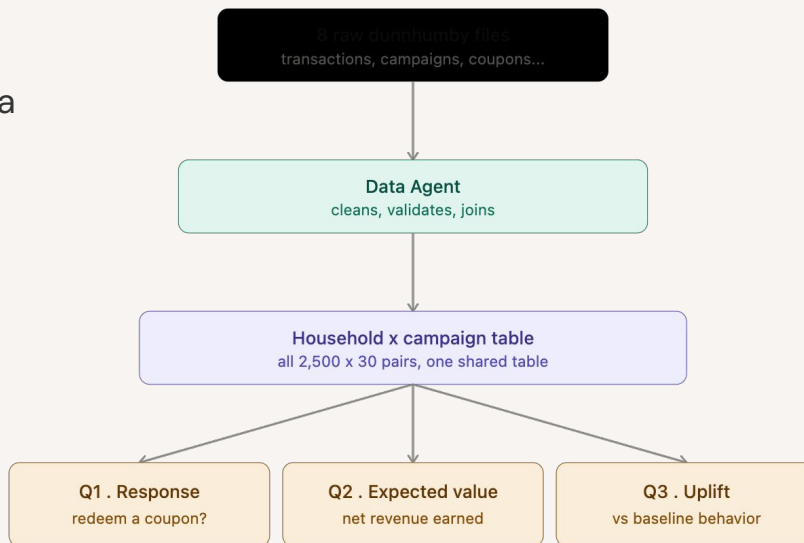
Agent Architecture

Design one shared agent pipeline that serves all three models, where only each agent's instructions change per model.

Data Agent – what will it produce?

Data Confirmed

- 7,208 household-campaign mailings, 889 with a redemption (12.33%) – matches the proposal exactly
- Redemption rate by campaign type: TypeA 16.0%, TypeB 7.9%, TypeC 7.7%



Data Agent – Progress

- First-draft ingestion script built and run successfully on all 8 real files (2.6M transaction rows, 36.8M display/mailer rows)
- Produces clean tables + a data quality report
- Repo scaffolded with a README and folder structure matching the five-agent pipeline
- Verified against the real data, not just the proposal, catching three issues that shaped how Models 2 and 3 define their targets

Findings from validating the data — and how they shaped the models

- **Net revenue formula was backwards.** The proposal's formula added the coupon discount back instead of subtracting it. Resolved: Model 2 uses SALES_VALUE directly, since the discount is already reflected in it.
- **TypeA coupon targeting is untraceable.** Each household gets 16 coupons from a pool of ~200, and which 16 isn't recorded. Resolved: Model 3 scopes to TypeB/TypeC only, where every household gets every coupon.
- **Campaigns overlap heavily and aren't in date order (74 of 435 pairs overlap).** This is why train/test splits need to go by date, not campaign ID.

Model 1: Campaign Response Probability

What is the probability that the household will redeem a coupon from a campaign?

- Output: number (0% – 100%); e.g. *household 123 + campaign 22 = 34%*
- Type of problem: binary classification

Household

- 2,500 households
- 2 years (711 days)
- Demographics for 801 / 2500 households (32%)

Campaigns

- Type A – personalized, larger
- Type B – same small set
- Type C – same small set

Model 1: Campaigns

	Type A	Type B	Type C
How coupons are assigned	Personalized, 16 coupons picked from large pool	Everyone in campaign gets all of the campaign's coupons	Everyone in campaign gets all of the campaign's coupons
Coupons per campaign	181 to 209 in the pool (household sees 16, about 8%)	2 to 33	1 to 34
Number of campaigns	5	19	6
Typical length (days)	41 to 56 (median 48)	33 to 62 (median 33)	33 to 162 (median 65)
Mailings (household x campaign)	3,979 (55% of all)	2,655 (37%)	574 (8%)
Mailings per campaign (avg)	about 800	about 140	about 95
Redemption rate	16.0% (635 of 3,979)	7.9% (210 of 2,655)	7.7% (44 of 574)
Share of all redeemers	71%	24%	5%

Which files does Model 1 need?

File	Rows	Role in Model 1	Needed?
campaign_table	7,208	Defines rows: who was mailed which campaign	Required
campaign_desc	30	Campaign type, start and end day (the time cutoff)	Required
coupon_redempt	2,318	Defines label: did the household redeem?	Required
transaction_data	2,595,732	Purchase history, the main source of features	Required
product	92,353	Product categories for category and brand features	Required
coupon	124,548	Which products each campaign's coupons cover	Required
hh_demographic	801	Demographic features for the 32% who have them	Optional
causal_data	36,786,524	In-store displays and weekly mailer features	Not for v1

Model 2: Expected Financial Value

- Question:
 - If a household redeems from a campaign, how much financial value do we expect from campaign-related products during the campaign period?
- Target:
 - Sum of **SALES_VALUE** for campaign-related products during **START_DAY-END_DAY**
 - **SALES_VALUE** is used directly because retailer coupon-match discounts are already reflected in the field
- Predictive Model:
 - Regression on redeemed household × campaign pairs

Model 2: Defining Campaign-Related Value

- Type B/C only
 - 254 redeemed pairs
 - Known coupon exposure, cleaner campaign-related product target
- Type A
 - Exact 16 coupons unobserved
 - Household-level product exposure uncertain
 - Open methodological question: Investigate alternative ways to operationalize the target for Type A

Model 3: True Incremental Uplift

Isolating Direct-Mail Impact from Baseline Habit & Store Confounders

Model Objective

Estimate the *incremental* effect of mailing a campaign i.e. how much more likely a household is to buy eligible products **because of the mailer**, beyond their organic baseline habit (what they'd buy anyway) or response to in-store promotions.

Observation Level

One household × one campaign — scoped to TypeB/TypeC Campaigns

Scope Rationale: Why Exclude Type A?

Type B/C: Households receive every coupon in the campaign.

Type A: Households receive only 16 untracked coupons from a pool of 17k–35k products. Testing full pools as "eligible" caused a **96% False-Positive** conversion rate in trial testing.

25 / 30
Campaigns

3,229
Mailings (45%)

Target / Outcome

Binary Metric (1 / 0)

Did the household purchase an eligible product (**PRODUCT_IDs**) during the active campaign window?

Strict Exclusion Filter

Excludes purchases occurring while the product was concurrently featured on an **in-store displays**.

Model 3: Data Feasibility



Treatment Group

Mailed this campaign

Households mapped in campaign table who received the direct mailer.

Verified Against: `campaign_table`



Control Group

Not mailed this campaign

Baseline households not in campaign table, did not receive this or any overlapping campaign.

Verified Against: `campaign_table` + `campaign_desc`



Confounder Filter

In-store promotion filter

Flags purchases driven by in store displays via shared `PRODUCT_ID`, `STORE_ID`, & `WEEK_NO` (same product, floor & week).

Verified Against

`causal_data` → join to `transaction_data`



Clean Causal Data

Isolated Impact

Successfully isolates direct-mail influence from baseline organic habit and store-level promotion confounders.

Ready for Uplift Modeling



Limitation: Campaign Targeting is Non-Random

Treated households spend **~4.6x more** pre-campaign than control households. This design is framed as *uplift modeling adjusting for observed confounders*, rather than a pure randomized-experiment causal estimate.

Model 3: ML Methodology & Evaluation

Execution Strategy and System Design

Methodology: T-Learner (2 Learner) Architecture

Approach: Model T (treated households) vs. Model C (control households) trained independently.

$$\text{Uplift} = P(\text{buy} \mid \text{Treatment}) - P(\text{buy} \mid \text{Control})$$

Rationale: Prevents single models (S-Learner) from ignoring the treatment flag when dominated by heavy baseline features like past spend.

Evaluation Metric: Qini Curve

Uplift Cumulative Gain

Standard ROC-AUC is invalid for causal uplift modeling because real-world counterfactuals cannot be observed directly.

 Evaluates true incremental customer conversion

Key Business Finding: Value-Add over Propensity Models (Model 1)

"Sure Things" Flaw

Current campaigns target heaviest most loyal spenders who buy anyway. Model 1 (Propensity) misallocates budget to them because overall purchase probability is high.

"Persuadables" Opportunity

Untargeted households can exhibit the largest response gap between "mailed" and "not mailed". Uplift modeling isolates true incremental impact to drive real revenue.

Agent Architecture

One agent architecture serves all three prediction questions. Each agent does the same job in the same way for every model. Only the instructions it receives change with the model.

EACH AGENT IS SHARED, FOR ITS OWN REASON

1. Data	All three models use the same data files, so the cleaning rules are the same.
2. Feature	One shared table of household features serves all three; each model keeps only its own campaigns, rows and target.
3. Modeling	Same steps (reshape the data, train, tune); the instructions say which kind of model to build.
4. Evaluation	Same steps (test on later campaigns, compare with a baseline); the instructions say how to measure success.
5. Scoring	Runs any approved model the same way; each model adds one score column.
6. Decision Lab	One application; each score is a column it can rank by or add up.
7. Decision Support	Same tools to look up and explain results; it is told what each score means.

Model 3 plugs its own method (two models: mailed and not mailed) and its own evaluation (Qini curve) into the shared Modeling and Evaluation agents.

Agent Architecture: Data and Feature Agents

1 · DATA AGENT

Does: cleans and checks all 8 files, and checks that they agree with each other.

Instructions: one cleaning rulebook for the whole dataset (expected columns, checks, and whether to fix, flag or stop). The same for every model, because the data is the same.

2 · FEATURE AGENT

2a, shared: builds one table of 75,000 rows, one per household and campaign ($2,500 \times 30$). Columns: features from before each campaign, and every model's target, including Model 3's purchase outcome without display-driven purchases.

2b, per model: the instructions say which campaigns, rows and target column to keep.

Model	Campaigns	Rows	Target column
1. Response	All 30	Mailed: 7,208	Redeemed (yes/no)
2. Expected value	All 30 (or TypeB/C)	Redeemed: 889 (TypeB/C: 254)	Spend on campaign products (\$)
3. Uplift	TypeB/C only (25)	Mailed and not mailed: 43,396	Bought a campaign product (yes/no)

Agent Architecture: Modeling and Evaluation Agents

3 · MODELING AGENT

3a, reshape the table for each model type: logistic regression needs categories as yes/no columns and numbers on a similar scale; a boosting model uses the table almost as it is. Same information, different format.

3b, apply the model: train and tune on the earlier campaigns.

4 · EVALUATION AGENT

Does: tests the models on the later campaigns, compares them with a baseline, and decides: approve, or go back to features or modeling.

Model	Modeling instructions	Evaluation instructions
1. Response	Yes/no models (e.g. logistic regression, LightGBM)	Redeemers caught in the top 20%, vs. "redeemed before"
2. Expected value	Dollar models (e.g. linear regression)	Average dollar error, vs. usual spend
3. Uplift	Two models (mailed, not mailed); uplift = difference	Qini curve, vs. random targeting

5–7 · Scoring, Decision Lab, Decision Support: shared; each model adds one score column (Model 1 × Model 2 = expected value).